Learning goals:

- Understand oscillatory signals and their spatial and spectral components;
- Understand the difference between evoked responses (e.g., ERPs) and induced (e.g., ERD) responses;
- Understand important signal analysis and (pre-)processing concepts like sampling frequency, Nyquist frequency, aliasing, detrending, spectral filtering, convolution, Hilbert transform, FFT, and spectrogram;
- Understand the spectrum and spectrogram;
- Understand the envelope of a signal, particularly thr Hilbert transform.

Question 1: Refresh you understanding

Refresh your understanding in the following concepts:

- Convolution: Wikipedia, Mike X Cohen
- Shannon-Nyquist sampling theorem: Wikipedia, Steve Brunton
- Aliasing: Wikipedia
- Fourier transform: 3Blue1Brown
- Fourier series: 3Blue1Brown
- Time-frequency representation: Wikipedia, Mike X Cohen
- Envelope: Wikipedia, Hilbert transform, Mike X Cohen

Question 2: Convolution

(a) Explain in your own words what *convolution* is.

(b) How is convolution used in spectral filters (e.g., a bandpass)?

Question 3: Sampling theorem

(a) What does a *sampling frequency* tell you?

(b) What is the *Nyquist* frequency?

(c) What is *aliasing*?

(d) How can aliasing be prevented?

Question 4: Filters

(a) What is a *spatial* filter? Briefly describe a typical use-case.

(b) What is a *spectral* filter? Briefly describe a typical use-case.

(c) What are highpass, lowpass, bandpass, bandstop, and notch filters? Provide a use-case for each.

Question 5: Jupyter notebook oscillatory signals

Now put the theory into practice by going through the `oscillatory signals` Jupyter notebook.

Question 6: Submitting to Brightspace

For this assignment, please submit the following:

- The Jupyter notebook with your solutions
- A report (pdf extension) with the answers to the theoretical questions. In this report, add a line stating who contributed to what (please, adhere to the code of conduct). Additionally, please add a line on whether, where, and how you used GenAI for this assignment.

Question 7: (Optional) Test your understanding with real exam questions

(a) Oscillatory signals

- 2020-2021 resit Part 5
- 2021-2022 Part 5
- 2024-2025 resit Part 4